Label-Efficient Dark-Vessel Detection in SAR:

Does In-Domain Self-Supervision Beat Optical Transfer?

John Roth & Kyle Wagner · EN.705.643 Deep Learning Developments with PyTorch

1 Problem and Motivation

Detecting "dark" vessels — ships not broadcasting AIS — in spaceborne synthetic aperture radar (SAR) is the operationally critical, label-starved case in maritime monitoring. The xView3-SAR benchmark provides Sentinel-1 imagery for this task, but its scenes are few and its dark-vessel labels are precious and human-verified, so supervised-only training risks overfitting. Pretrained foundation backbones promise to close that gap, but it is unclear which kind transfers best to this regime: a generic optical-satellite model, a SAR-specific one, or labeled out-of-domain SAR. We measure this directly, holding the detector and its size fixed and comparing downloaded pretrained backbones, so the only variable is the initialization. We pretrain nothing ourselves — a deliberate choice that removes pretraining compute and novel pretraining code as risks, and concentrates effort on a clean, metric-matched evaluation.

2 Research Question

Under one fixed-size detection harness, how much labeled xView3 data does each class of pretrained backbone save for dark-vessel detection — and does a SAR-domain foundation model (SARMAE) deliver materially better label efficiency than a strong optical one (SatDINO) in the scarce-label regime? **Hypothesis:** the SAR-domain foundation model yields the steepest label-efficiency gains and the largest dark-vessel-recall advantage at low budgets, with arms ordered random $<$ SatDINO $\lesssim$ supervised $<$ SARMAE at low label fractions and converging as labels grow; the optical$\rightarrow$SAR gap quantifies the value of domain-matched pretraining.

3 Approach

The harness is a plain ViT-B/16 backbone ($\sim$86M parameters) with a heatmap detection head: each vessel is a Gaussian blob on a stride-4 response map, decoded by peak-finding with distance-based non-maximum suppression. This is point-native: xView3's labels are points and its official metric matches predictions to ground truth by distance (not box IoU), so a heatmap-and-distance formulation mirrors the metric exactly and avoids synthesizing bounding boxes the data never had. **Fairness by construction:** all four study arms use the same ViT-B/16 architecture, the same head, the same fine-tuning schedule and seeds, and a single fixed input representation ([VH, VV, VH$-$VV] in dB); only the downloaded initialization differs. Both foundation-model arms (SatDINO, SARMAE) are ViT-B/16 and both fine-tune end to end, so there is no architecture or size confound between them; they do differ in self-supervised method (DINO vs. masked autoencoding), which we treat as a documented caveat rather than control, comparing the best released foundation model of each domain as-is. We report parameter count and GPU-hours per arm. Data is xView3 via the SARFish mirror (Sentinel-1 GRD, dual-pol VH/VV, scene-level splits); LS-SSDD-v1.0 provides the labeled SAR-ship source for the supervised-transfer arm.

Arm	Pretraining	Role
Controlled study (plotted on the label-efficiency curve)		
1	none (random init)	no-pretraining floor
2	SatDINO ViT-B (fMoW-RGB, optical DINO)	optical-domain transfer anchor
3	SARMAE ViT-B (SAR-1M, downloaded)	SAR-domain FM; the central comparison
4	supervised on LS-SSDD-v1.0 (we train)	labeled near-domain transfer
Reported separately (not on the curve)		
5	best FM + pooled supervised, unconstrained	leaderboard entry
6	YOLO26-COCO; LocateAnything zero-shot	external reference range

SatDINO (Arm 2) is an optical-satellite foundation model (DINO self-distillation on fMoW-RGB; Straka & Gruber 2025); SARMAE (Arm 3) is a SAR foundation model — a masked autoencoder with speckle-aware enhancement, pretrained on the million-image SAR-1M dataset (Liu et al., CVPR 2026). Both are downloaded ViT-B/16 checkpoints; we pretrain neither. Two honest caveats, stated rather than hidden. First, the two arms differ in self-supervised method (DINO vs. MAE) as well as domain — no downloadable optical ViT-B MAE was available to match SARMAE's method, so we compare the strongest released model of each domain as-is. Second, each was pretrained on 3-channel inputs that are not dual-pol SAR (SatDINO on optical RGB, SARMAE on single-pol SAR mapped to three channels), so feeding both the fixed [VH, VV, VH$-$VV] tensor leaves a third channel each adapts during fine-tuning — a bounded transfer gap shared symmetrically by the two foundation-model arms, not a confound between them. This situates the work as a controlled label-efficiency benchmark of pretrained backbones: it isolates optical-domain versus SAR-domain foundation-model transfer, specifically on dark-vessel, scarce-label detection, which to our knowledge has not been done, and it connects to the pre-training-versus-self-training question of Zoph et al. Arm 5 stacks the best foundation model with pooled supervised data and unconstrained tricks (ensembling, test-time augmentation, pseudo-labeling), reported separately as a leaderboard attempt, not part of the controlled comparison. All training uses one PyTorch Lightning code path, so seeding, mixed precision, and distributed training are identical across arms.

4 Evaluation Plan

We split xView3 training scenes 75/15/10 at the *scene* level (no chip-level leakage); the ∼50 human-verified validation scenes are reserved and touched once, at final evaluation. The headline deliverable is a **label-efficiency curve**: detection F1 versus 10/25/50/100% of real labels, one line per study arm (Figure 1). The gap above the floor at 10% labels quantifies the value of a pretrained initialization, and the arm ordering doubles as an optical→SAR transferability result (we expect SatDINO < SARMAE if domain match predicts transfer). We additionally report *dark-vessel recall* and *close-to-shore F1*, the operationally hardest slices, where any SAR-domain advantage should concentrate.

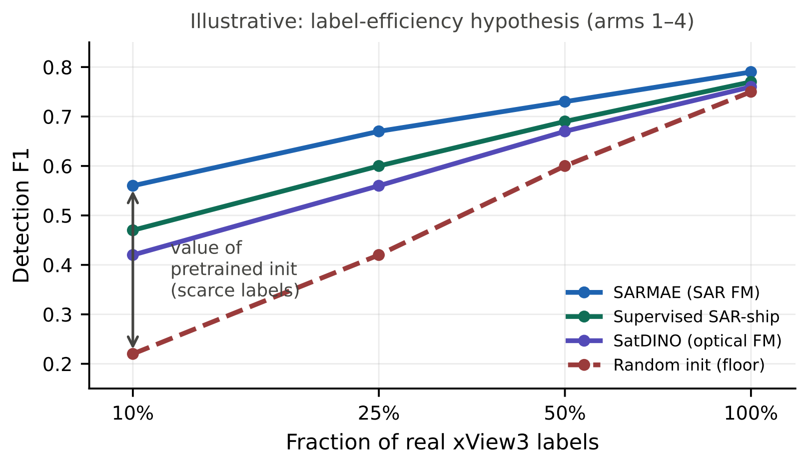

Figure 1: Hypothesized label-efficiency curves for the four study arms (illustrative). Pretraining is expected to help most when labels are scarce; the arms converge as labels grow.

5 Timeline, Compute, and Feasibility

Two contributors work largely independent lanes — one owns data and the scorer, the other the detection harness and evaluation — with no hard dependency, since both foundation models are downloaded and there is no pretraining handoff to block on. Compute is one 8×V100 (32 GB) node plus an RTX 5070 Ti for development. Because we pretrain nothing, the total is modest: each fine-tune is a few GPU-hours and the full ∼36-run grid plus the single supervised-transfer backbone comes to roughly **130 GPU-hours**, about a day on the node — an order of magnitude below what self-pretraining a SAR model would cost, which is precisely why we download instead. The foundation-model arms run *first*, because the main technical risk is the channel-format gap (whether a patch-embed pretrained on 3-channel optical or single-pol SAR accepts the [VH, VV, VH−VV] input); a quick load-and-train check de-risks the shared input for all arms before the grid. We will confirm each dataset and checkpoint downloads and opens before committing. As prior work just like this — foundation models and label-efficient detection for SAR — appears at venues such as the CVPR EarthVision and ICCV remote-sensing workshops, we will target one such workshop for submission.

Weeks	Milestone
1–2	SARFish/LS-SSDD download, chipping, sacred scorer + harness
2–4	Run SatDINO + SARMAE (downloaded) + floor arms; channel-format check; references
4–6	Supervised arm; full label-fraction grid; seed reruns
6–8	Corpus ablation; dark-vessel and shoreline error analysis
8+	Writeup; optional unconstrained challenge entry

Key References

[1] Paolo et al. *xView3-SAR: Detecting Dark Fishing Activity Using SAR Imagery.* NeurIPS 2022.
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[3] Straka & Gruber. *SatDINO: A Deep Dive into Self-Supervised Pretraining for Remote Sensing.* 2025 (arXiv:2508.21402).
[4] Liu et al. *SARMAE: Masked Autoencoder for SAR Representation Learning.* CVPR 2026 (arXiv:2512.16635).
[5] Zhang et al. *LS-SSDD-v1.0: A Large-Scale SAR Ship Detection Dataset.* 2020.
[6] Zoph et al. *Rethinking Pre-training and Self-training.* NeurIPS 2020.
[7] Jocher et al. *Ultralytics YOLO26*, 2026.